

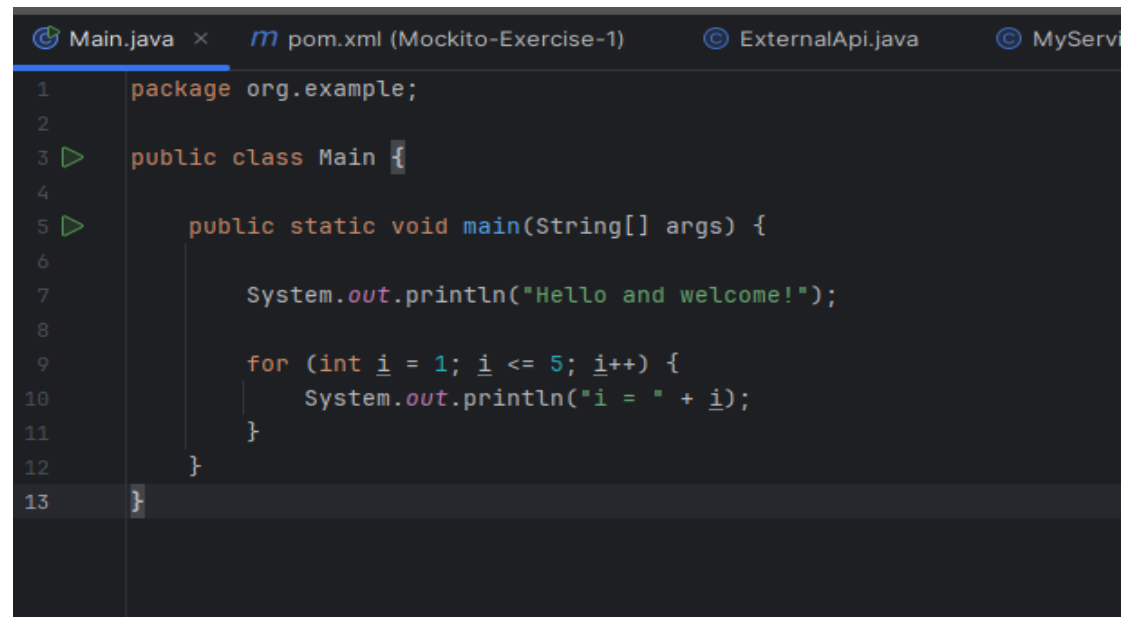
# TDD using JUnit5 and Mockito

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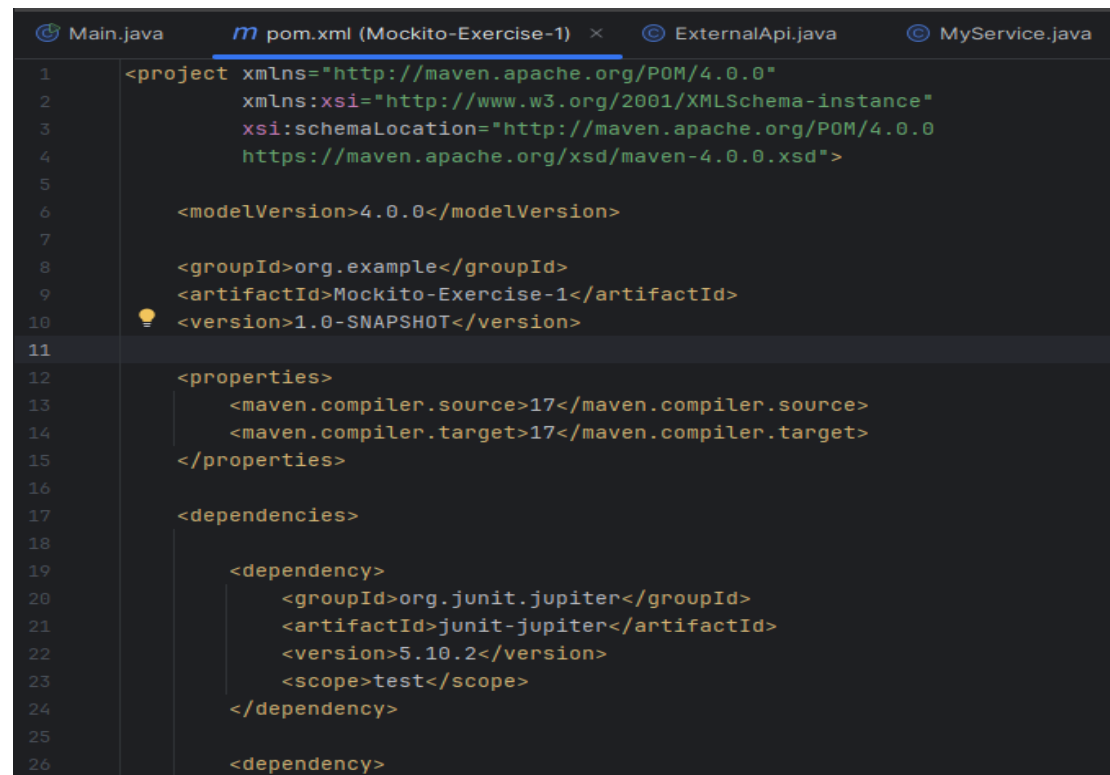
## Exercise 1:Mocking and Stubbing:

### Main.java:

A screenshot of an IDE showing the Main.java file. The code is as follows:

```
1 package org.example;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         System.out.println("Hello and welcome!");
8
9         for (int i = 1; i <= 5; i++) {
10             System.out.println("i = " + i);
11         }
12     }
13 }
```

### Pom.xml:

A screenshot of an IDE showing the pom.xml file. The code is as follows:

```
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4                             https://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6     <modelVersion>4.0.0</modelVersion>
7
8     <groupId>org.example</groupId>
9     <artifactId>Mockito-Exercise-1</artifactId>
10    <version>1.0-SNAPSHOT</version>
11
12    <properties>
13        <maven.compiler.source>17</maven.compiler.source>
14        <maven.compiler.target>17</maven.compiler.target>
15    </properties>
16
17    <dependencies>
18
19        <dependency>
20            <groupId>org.junit.jupiter</groupId>
21            <artifactId>junit-jupiter</artifactId>
22            <version>5.10.2</version>
23            <scope>test</scope>
24        </dependency>
25
26        <dependency>
```

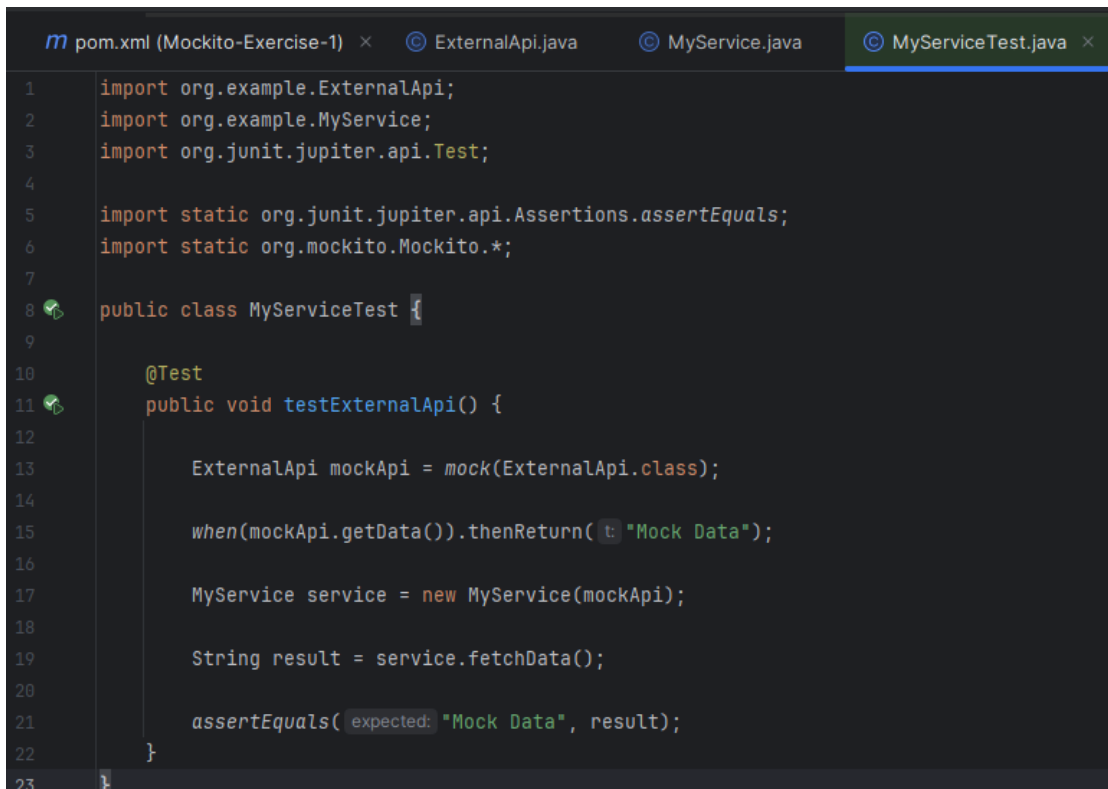
### ExternalAPI.java:

```
Main.java  pom.xml (Mockito-Exercise-1)  ExternalApi.java x
1 package org.example;
2
3 public interface ExternalApi { no usages
4     String getData(); no usages
5 }
```

### My Service.java:

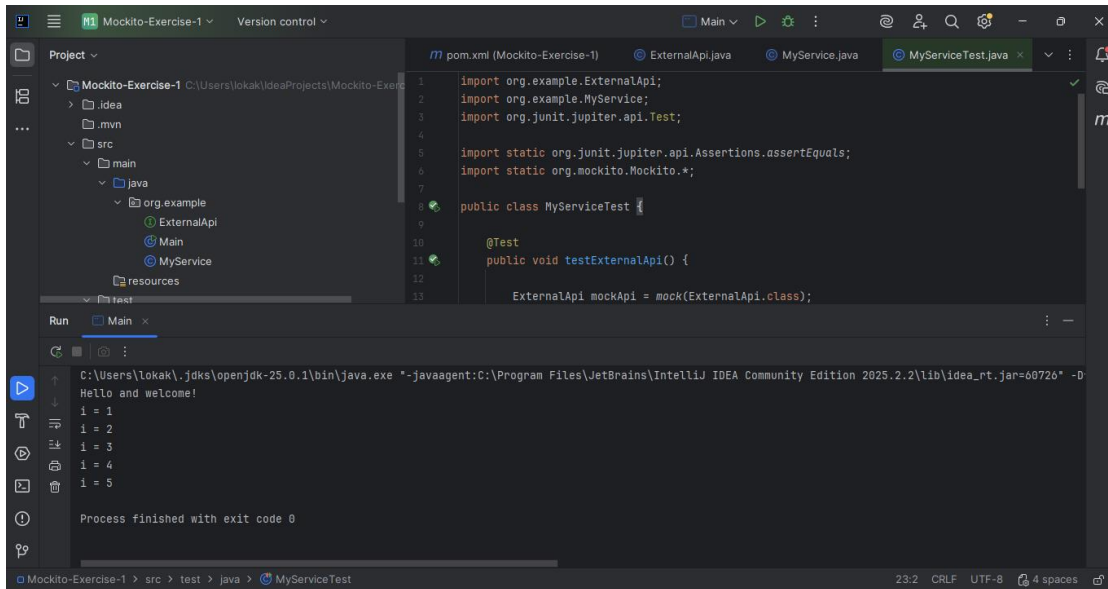
```
pom.xml (Mockito-Exercise-1) x  ExternalApi.java  MyService.java x
1 package org.example;
2
3 public class MyService { no usages
4
5     private ExternalApi externalApi; 2 usages
6
7     public MyService(ExternalApi externalApi) { no usages
8         this.externalApi = externalApi;
9     }
10
11     public String fetchData() { no usages
12         return externalApi.getData();
13     }
14 }
```

## MyServiceTest.java:



```
1 import org.example.ExternalApi;
2 import org.example.MyService;
3 import org.junit.jupiter.api.Test;
4
5 import static org.junit.jupiter.api.Assertions.assertEquals;
6 import static org.mockito.Mockito.*;
7
8 public class MyServiceTest {
9
10     @Test
11     public void testExternalApi() {
12
13         ExternalApi mockApi = mock(ExternalApi.class);
14
15         when(mockApi.getData()).thenReturn("Mock Data");
16
17         MyService service = new MyService(mockApi);
18
19         String result = service.fetchData();
20
21         assertEquals("Mock Data", result);
22     }
23 }
```

## Output Screenshot:

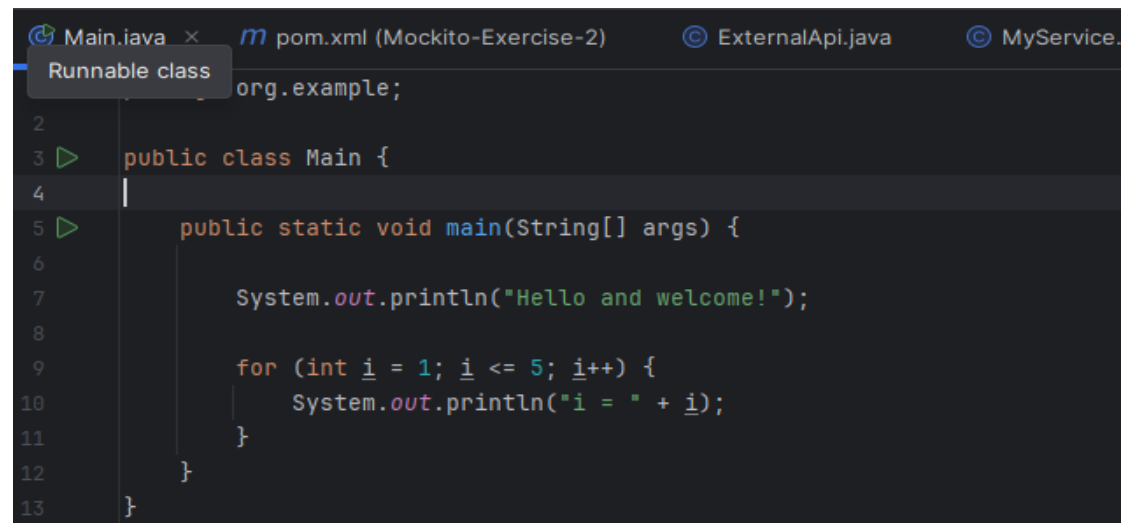


```
C:\Users\lokak\jdk\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2.2\lib\idea_rt.jar=60726" -D
Hello and welcome!
1 = 1
1 = 2
1 = 3
1 = 4
1 = 5
Process finished with exit code 0
```

Mockito-Exercise-1 > src > test > java > MyServiceTest 23:2 CRLF UTF-8 4 spaces

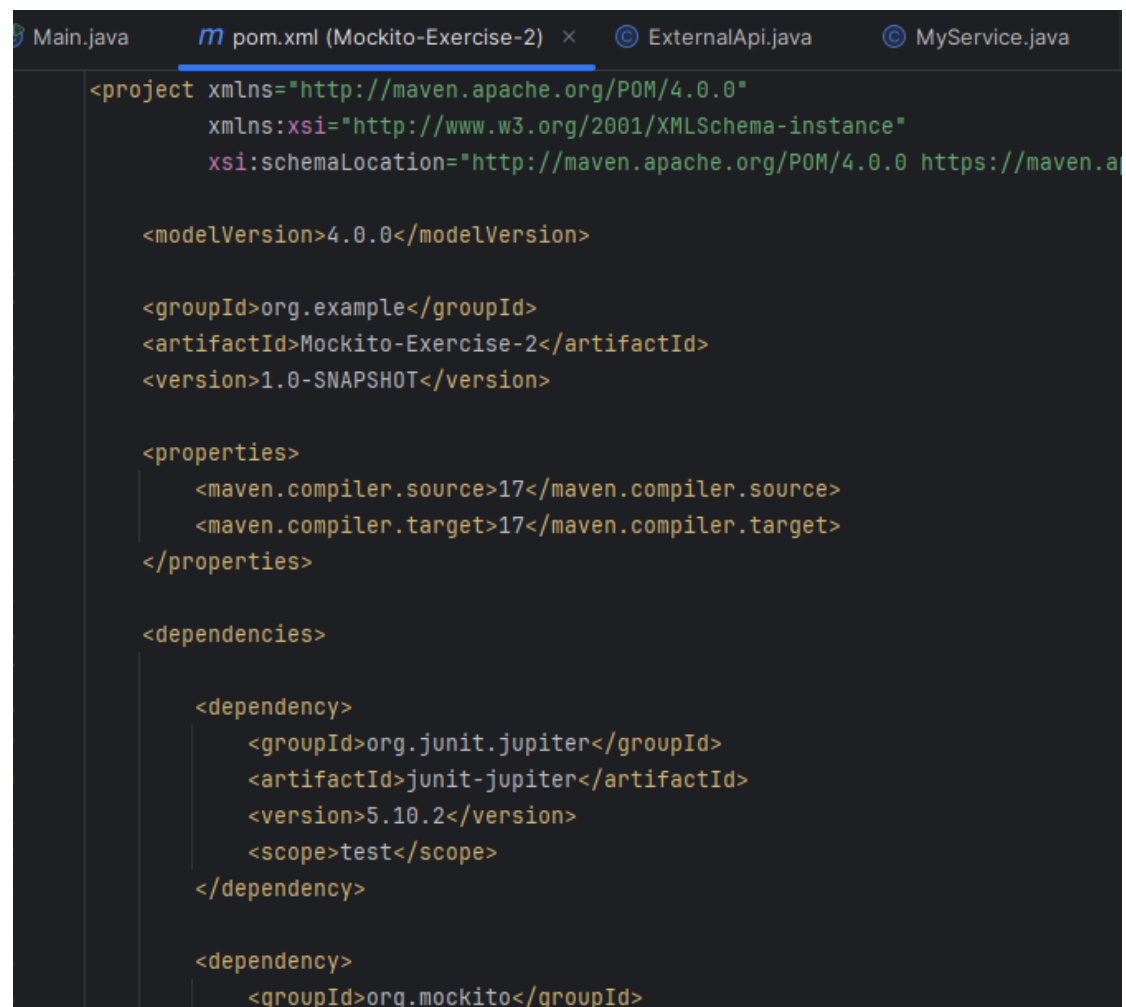
## Exercise 2: Verifying Interactions:

### Main.java:



```
1 org.example;  
2  
3 public class Main {  
4  
5     public static void main(String[] args) {  
6  
7         System.out.println("Hello and welcome!");  
8  
9         for (int i = 1; i <= 5; i++) {  
10             System.out.println("i = " + i);  
11         }  
12     }  
13 }
```

### Pom.xml:



```
1 <?xml version="1.0" encoding="UTF-8"?>  
2 <project xmlns="http://maven.apache.org/POM/4.0.0"  
3     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
4     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/maven-v4_0_0.xsd">  
5  
6     <modelVersion>4.0.0</modelVersion>  
7  
8     <groupId>org.example</groupId>  
9     <artifactId>Mockito-Exercise-2</artifactId>  
10    <version>1.0-SNAPSHOT</version>  
11  
12    <properties>  
13        <maven.compiler.source>17</maven.compiler.source>  
14        <maven.compiler.target>17</maven.compiler.target>  
15    </properties>  
16  
17    <dependencies>  
18        <dependency>  
19            <groupId>org.junit.jupiter</groupId>  
20            <artifactId>junit-jupiter</artifactId>  
21            <version>5.10.2</version>  
22            <scope>test</scope>  
23        </dependency>  
24  
25        <dependency>  
26            <groupId>org.mockito</groupId>
```

## ExternalApi.java:

```
Main.java x pom.xml (Mockito-Exercise-2) ExternalApi.java x MyServ
1 package org.example;
2
3 public interface ExternalApi { no usages
4     String getData(); no usages
5 }
```

## MyService.java:

```
xml (Mockito-Exercise-2) x ExternalApi.java MyService.java x MyServiceVerificat
1 package org.example;
2
3 public class MyService { no usages
4
5     private ExternalApi externalApi; 2 usages
6
7     public MyService(ExternalApi externalApi) { no usages
8         this.externalApi = externalApi;
9     }
10
11     public String fetchData() { no usages
12         return externalApi.getData();
13     }
14 }
```

## MyServiceVerificationTest.java:

```
xml (Mockito-Exercise-2) ExternalApi.java MyService.java MyServiceVerificationTest.java x
1 import org.example.ExternalApi;
2 import org.example.MyService;
3 import org.junit.jupiter.api.Test;
4
5 import static org.mockito.Mockito.*;
6
7 public class MyServiceVerificationTest {
8
9     @Test
10     public void testVerifyInteraction() {
11
12         ExternalApi mockApi = mock(ExternalApi.class);
13
14         MyService service = new MyService(mockApi);
15
16         service.fetchData();
17
18         verify(mockApi).getData();
19     }
20 }
```

## Output Screenshot:

